Nahyuk Lee

Position

Research Scientist
VUNO Inc.

Contact Information

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Birthday

Jan 13th, 2000.

Research Interests

Computer Vision & Machine Learning, in particular, the problems of 3D computer vision:

- Learning visual correspondences (*i.e.*, 2D & 3D feature matching) and its applications
- 3D point cloud registration & Geometric shape assembly
- Geometric deep learning (*i.e.*, equivariant learning)
- Representation learning for 3D generative model

Education

Pohang University of Science and Technology (POSTECH), Pohang, Korea
M.S. in Computer Science and Engineering **Feb 2023 - Feb 2025**

- Thesis: *Combinative Matching for Geometric Assembly*
- Advisor: Prof. Minsu Cho
- Admission with the highest distinction (**Bang Sung-Yang Fellowship**)

Chung-Ang University, Seoul, Korea
B.S. in Computer Science and Engineering **Mar 2019 - Feb 2023**

- Advisor: Prof. Junseok Kwon
- Cumulative GPA: 4.31/4.5 (Summa Cum Laude, **1st in Class of 2023**)

Hyundai Chung-Un High School, Ulsan, Korea
Natural Science Track **Mar 2015 - Feb 2018**

Experience

Computer Vision and Geometry Group, ETH Zurich, Zurich, Switzerland (Remote)
Research Collaborator **Oct 2025 - now**

- Working with Dr. Sunghwan Hong and Prof. Marc Pollefeys
- Co-working with master's student in ETH Zurich (Zhiang Chen), focusing on topology-aware representation alignment for rectified point flow.
  — *Representation alignment*: Proposed TORA, a framework that distills relational structure from frozen pretrained 3D encoders (*e.g.*, Uni3D) into a flow-matching backbone via token-wise cosine matching and Centered Kernel Alignment (CKA), achieving faster convergence (up to **6.9**×), improved in-distribution accuracy, and strong zero-shot transfer [arXiv 2026].

VUNO Inc., Seoul, Korea
Research Scientist (Alternative Military Service) **Jan 2025 - now**

- Working with Dr. Sunghoon Joo (R&D Lead).
- **Research**: Researching 3D computer vision for lung CT and brain MRI analysis.
  — *Geometric deep learning*: Researching SO(3)-equivariant brain lesion segmentation for computer-assisted diagnosis, collaborating with Asan Medical Center. (Jun 2025 - now)
  — *Model quantization*: Researching robust Quantization-Aware Training (QAT) methods for group-equivariant networks. (Nov 2025 - Feb 2026)
  — *Visual correspondence*: Researched efficient point-based deformable lung CT registration, achieving SoTA on the Lung250-M benchmark [Appl. Sci. 2025]. (Mar - May 2025)

- **Product**: Led key components in VUNO Med®-DeepBrain® and VUNO Med®-LungCT AI™.
  — *DeepBrain*: Developed LLM-powered (*e.g.*, GPT-OSS, MedGemma) medical report-generation system to automate radiology workflows. (Oct - Dec 2025)
  — *LungCT*: Developed and optimized volumetric super-resolution and solid-portion segmentation modules, achieving a **2.66**× inference speed-up while reproducing the legacy performance. (Jan - May 2025)

Computer Vision Lab, POSTECH, Pohang, Korea
Graduate Research Assistant **Jul 2022 - Dec 2024**

- Advised by Prof. Minsu Cho
- Researched problems in 3D point cloud matching, with a particular focus on point cloud registration, geometric shape assembly, and equivariant learning.
  — *Robust point cloud matching*: Proposed an SE(3)-equivariant ***combinative*** matching methodology, resolving local correspondence ambiguity by jointly modeling geometric similarity and complementary occupancy cues [**ICCV 2025 Highlight**].
  — *Efficient point cloud matching*: Proposed efficient approximation of high-order convolution, demonstrating SoTA performance on geometric shape assembly while requiring **21.5**× fewer FLOPs and **3.38**× fewer parameters than prior SoTA [**ICML 2024**, IPIU 2024 Best Paper].
- Collaborated with VGI Lab, SNU (Kanghee Lee, Prof. Jaesik Park).

Undergraduate Research Fellowship **Jan 2022 - Feb 2022**

- Advised by Prof. Jaesik Park
- Researched point cloud augmentation and sampling methods for robust 3D shape classification.
- Collaborated with a student at CUHK (Hansheng Guo).

Computer Vision Lab, Korea University, Seoul, Korea
Research Intern (full-time) **Mar 2021 - Nov 2021**

- Advised by Prof. Sangpil Kim
- Researched audio-guided image manipulation [NeurIPSW 2021], and event-based photometric stereo.

Computer Graphics Lab, Chung-Ang University, Seoul, Korea
Research Intern (full-time) **Dec 2019 - Feb 2021**

- Advised by Prof. Sanghyun Seo and Prof. Kyunghyun Yoon
- Researched automated 3D reconstruction pipeline [JDCS 2020], and computer vision-based affective computing [CSCI 2020, JDCS 2021].

Publications

My research has been published in top-tier computer vision and machine learning venues, including **ICCV** and **ICML**, with additional works in applied research on medical AI and interdisciplinary venues (All papers are available on Google Scholar or my Homepage.). * denotes equal contribution.

- P: Preprint, C: Conference, W: Workshop, J: Journal

[P1] **Nahyuk Lee***, Zhiang Chen*, Marc Pollefeys, Sunghwan Hong, "TORA : Topological Representation Alignment for 3D Shape Assembly," arXiv 2026

[C4] **Nahyuk Lee***, Juhong Min*, Junhong Lee, Chunghyun Park, Minsu Cho, "Combinative Matching for Geometric Shape Assembly," *in Proceeding of the IEEE/CVF International Conference on Computer Vision* **(ICCV)**, 2025, **Highlight** ($263/11,152 \approx$ **2.35**%)

[C3] Doohyun Park, **Nahyuk Lee**, Sungjoo Lim, "Attention-based Interpretable Deep Learning with Radiomic Features for Pulmonary Nodule Classification," *in Medical Imaging and Deep Learning* (MIDL), Short Paper, 2025

[C2] **Nahyuk Lee***, Juhong Min*, Junha Lee, Seungwook Kim, Kanghee Lee, Jaesik Park, Minsu Cho, "3D Geometric Shape Assembly via Efficient Point Cloud Matching," *in Proceedings of the International Conference on Machine Learning* **(ICML)**, 2024

[C1] Taemin Lee, **Nahyuk Lee**, Sanghyun Seo, Dongwann Kang, "A Study on the Prediction of Emotion from Image by Time-flow depend on Color Analysis," in *Proceeding of the International Conference on Computational Science and Computational Intelligence* (CSCI), 2020

[W1] Seung Hyun Lee, **Nahyuk Lee**, Chanyoung Kim, Wonjeong Ryoo, Jinkyu Kim, Sang Ho Yoon, Sangpil Kim, "Audio-Guided Image Manipulation for Artistic Paintings," in *NeurIPS Workshop on Machine Learning for Creativity and Design* **(NeurIPS Workshop)**, 2021

[J3] **Nahyuk Lee**, Taemin Lee, "Bi-directional Point Flow Estimation with Multi-scale Attention for Deformable Lung CT Registration," in *Applied Sciences*, 2025

[J2] **Nahyuk Lee**, Taemin Lee, "The Study for Emotion Analysis from Images depend on Machine Learning Models," in *Journal of Digital Contents Society* (JDCS), 2021

[J1] **Nahyuk Lee**, Kyungtaek Lee, Youngsup Park, Sanghyun Seo, Taemin Lee, "Development of 3D Reconstruction and Object Recognition Model using Video," in *Journal of Digital Contents Society* (JDCS), 2020

Academic Services

Reviewer of international journals
Transaction on Pattern Analysis and Machine Intelligence (**TPAMI**) IF : 20.8

Reviewer of international conferences
European Conference on Computer Vision (**ECCV**)
International Conference on 3D Vision (**3DV**)
International Conference on Learning Representations (**ICLR**)
LSRW Workshop, Conference on Robot Learning (**CoRLW**)
WiCV Workshop, Asian Conference on Computer Vision (**ACCVW**)

Honors and Awards

Travel Grant, IEEE/CVF International Conference on Computer Vision (ICCV), 2025
- Waived registration fees with partial scholarship from ICCV 2025, **$1,100** (total)

18^{th} Kwanjeong Educational Foundation Fellowship, KEF, 2023
- Winner, received **$17,000** during 2 years of my graduate studies

Bang Sung-Yang Graduate Fellowship, POSTECH, 2023
- Admission with highest distinction, **$1,200**

16^{th} Kwanjeong Educational Foundation Fellowship, KEF, 2021
- Winner, received **$16,500** during 2 years of my undergraduate studies

University Innovation Support Project Scholarship, UISPC, 2021
- Research fellowship, **$750**

SW Specialization Scholarship, CAU, 2020 - 2022
- Research fellowship, **$4,200** (total)

Da Vinci Scholarship, CAU, 2019
- Admission with highest distinction, **$7,250**

Dean's List, CAU, 2019 - 2021
- 2019 Fall, 2020 Spring, 2020 Fall, 2021 Fall (4 times)

Awards
- Best Paper Award (**Grand Prize**, 1^{st} of 311 papers $\approx$ **0.3**%),
  36^{th} Workshop on Image Processing and Image Understanding (IPIU), 2024
- Excellence Prize in Research Report,
  Innovation Center for Engineering Education, CAU, 2022

- Excellence Prize in Project,
  Davinci Software Institute, CAU, 2021
- Best Paper Award,
  Digital Contents Society, 2020
- Excellence Award in Project,
  Innovation Center for Engineering Education, CAU, 2020
- Best Paper Award (top 10% papers),
  International Next-generation Convergence Technology Association (INCA), 2020
- Best Paper Award (2^{nd} Prize),
  Humanities Research Institute, CAU, 2020

Teaching Experience

Pohang University of Science and Technology (POSTECH)
Teaching Assistant **Feb 2024 - Mar 2024**
- CSED537/AIGS537 - Artificial Intelligence & Data Science (2024 Spring)

Ministry of National Defense, Republic of Korea
Instructor **Nov 2022 - Dec 2022**
- Conducted lectures on AI projects for South Korean soldiers and short-term service officers
- Collaborated with *Elice, Inc.*

Samsung Electronics Co., Ltd
Instructor **Aug 2022 - Oct 2022**
- Conducted lectures on data science and machine learning for newcomers (DS division)
- Collaborated with *Elice, Inc.*

Chung-Ang University
Teaching Assistant **Sep 2020 - Dec 2022**
- GE 56423 - Computational Thinking and AI Literacy (2022 Fall)
- GE 52431 - Understanding SW Programming for Problem Solving (2020 Fall, 2021 Spring/Fall, 2022 Spring)
- GE 51368 - Computational Thinking and Problem Solving (2021 Spring/Fall, 2021 Spring/Fall)
- CSE 17086 - Discrete Mathematics (2021 Fall)

Korea Scholar's Conference for Youth (KSCY)
Facilitator & Tutor **Jun 2019 - Feb 2021**
- Facilitator : 12^{th}-13^{th} KSCY computer engineering track camp facilitator
- Tutor : 15^{th} KSCY computer engineering track tutor

Leadership

Graduate School of Dept. of CSE, POSTECH
President of Student Council **Feb 2024 - Feb 2025**
- President of graduate school council of Department of Computer Science & Engineering, POSTECH.

Kwanjeong Educational Foundation Fellowship
Vice President **Jul 2023 - Jul 2024**
- Vice president of 18^{th} Kwanjeong Educational Foundation Fellowship mensches.

Chung-Ang University Artificial Intelligence Society (CUAI)
3^{rd} - 4^{th} President **Mar 2020 - Feb 2022**
- 3^{rd} (Mar 2020 - Feb 2021) and 4^{th} (Mar 2021 - Feb 2022) president of the student AI society at Chung-Ang University, CUAI.

INVITED PRESENTATIONS

Shaping Visual Intelligence: *The Past, Present, and Future of Computer Vision*, Kangwon National University, Samcheok, Korea, Nov. 2025

LLMs as Research Assistants: *A Practical Guide from My Latest Papers*, VUNO R&D Seminar, Seoul, Korea, Jun. 2025

3D Geometric Shape Assembly via Efficient Point Cloud Matching, Samsung AI Forum (**SAIF**) 2024, Suwon, Korea, Nov. 2024

Combinative Matching for Shape Assembly, Samsung Global Technology Symposium (**GTS**) 2023, Seoul, Korea, Mar. 2023

LANGUAGE SKILLS

Korean (native), English (fluent)

REFEREES

Prof. Minsu Cho (Academic Advisor)
Mu-Eun-Jae (無垠齋) Endowed Chair Professor
Associate Professor, Pohang University of Science and Technology (POSTECH)
e-mail : mscho@postech.ac.kr

Dr. Juhong Min (Coworker)
Senior Researcher, Samsung Research America (SRA)
e-mail : juhongm999@gmail.com

Dr. Sunghwan Hong (Coworker)
Post-doc Researcher, ETH Zürich
e-mail : hongsu@ethz.ch

Prof. Jaesik Park (Coworker)
Assistant Professor, Seoul National University
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